

PULSE: A Real-Time Wearable Multi-Sensor Telemetry and Affective Computing Framework via MQTT-WebSocket Bridging

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Project developed under the Samsung Innovation Campus (SIC) initiative

Abstract—Affective computing and real-time physiological monitoring play a critical role in next-generation Human-Computer Interaction (HCI), adaptive ambient computing, and stress detection. However, deploying multi-modal physiological sensing systems often suffers from network latency, sensor noise, baseline calibration drift, and state oscillation at affective classification boundaries. In this paper, we present PULSE (Predictive Ubiquitous Low-latency Sensor Evaluation), a lightweight, end-to-end framework designed for real-time biosignal telemetry and emotion sensing. PULSE ingests multi-modal data streams including Galvanic Skin Response (GSR), Photoplethysmography (PPG) heart metrics, and 6-axis MPU6050 Inertial Measurement Unit (IMU) kinematics transmitted over MQTT by an ESP32 wearable microcontroller. A Node.js middleware bridge transforms raw telemetry into WebSockets while running an affective classifier. The engine combines dynamic per-user baseline calibration, moving-window feature smoothing, weighted Z-score normalization, and a logistic squash mapping to derive an Arousal Index on a 0 to 100 scale. To mitigate rapid label flickering around decision boundaries, we introduce a dual-parameter hysteresis and dwell-time state machine. Finally, the framework couples affective state changes to ambient actuation, host-level audio gain and WebGL 3D spatial visualization. The ambient layer drives APA102 addressable lighting over hardware SPI, selected so that colour updates never suspend the interrupts required by time-critical sensor sampling, together with mood-indexed audio streamed from the wearable to a Bluetooth speaker over A2DP. Evaluation is simulation-based: a 30-session Monte Carlo study demonstrates robust noise rejection and realistic classification accuracy in the 62 to 91% range across affective bands, with the hysteresis layer trading a modest amount of instantaneous label accuracy for near-total elimination of spurious state oscillation. We further give an analytical latency budget, which shows that end-to-end responsiveness is bounded by the system’s own stability parameters and not by transport.

Index Terms—Affective Computing, Wearable Biosensors, Galvanic Skin Response, Photoplethysmography, MQTT, WebSockets, Hysteresis, Real-Time Telemetry.

I. INTRODUCTION

A. Motivation and Background

Affective computing, a field formally established by Picard [1], is concerned with systems that can recognize, interpret, and respond to human emotional and arousal states. As wearable sensing hardware has become smaller, cheaper, and more power-efficient, affective computing has moved from

controlled laboratory settings into everyday Human-Computer Interaction (HCI): adaptive user interfaces, ambient computing environments that respond to occupant stress, biofeedback-driven wellness applications, and gaming or educational platforms that adapt difficulty to a user’s engagement level. The common technical substrate underlying all of these applications is the same: a low-cost wearable must continuously sample one or more physiological signals, transmit them reliably to a processing unit, and convert noisy raw waveforms into a stable, interpretable affective label or continuous score.

Human affective states, ranging from calm and relaxation to heightened anxiety or high arousal, manifest through observable physiological phenomena [1]. Sympathetic nervous system activation triggers electrodermal activity (EDA/GSR) [2] and heart rate acceleration observable through photoplethysmography (PPG) [3], while physical motion dynamics measured by Inertial Measurement Units (IMUs) [7] supply situational context from which affective arousal can in principle be separated from simple physical exertion. In PULSE the IMU stream is acquired, smoothed and visualized, but it is not an input to the classifier evaluated here. Section III-C defines the composite score, states that exclusion, and gives the gating term that would lift it. Translating these multi-modal biosignals into accurate, responsive digital interactions requires overcoming three broad technical challenges, each of which directly motivated a design decision in PULSE.

B. Technical Challenges

1) **Heterogeneous Network Bridging:** Embedded microcontrollers such as the ESP32 [6] excel at low-latency, low-overhead publish/subscribe messaging using MQTT [4], a protocol purpose-built for constrained devices and unreliable links. Modern web clients and browser-based dashboards, however, do not natively speak MQTT; they rely on the WebSocket protocol [5] for bidirectional, full-duplex streaming of rich visual telemetry. Bridging these two ecosystems without introducing unacceptable latency or a single point of failure requires a middleware layer that can simultaneously act as an MQTT client and a WebSocket server, translating and fanning out messages between the two protocol domains in real time.

2) Inter-Individual Baseline Variance: Static, hard-coded thresholds on raw sensor values fail in practice because resting physiology varies substantially across individuals. Baseline skin resistance, resting heart rate, and even sensor placement pressure differ from person to person and from session to session. A value that indicates high arousal for one subject may be well within the normal resting range for another. This motivates a per-subject calibration phase that establishes an individualized statistical baseline before any classification is attempted.

3) Signal Oscillations and Decision Boundary Flickering: Sensor noise, motion artifacts, and transient micro-arousals (a brief startle, a cough, a shift in posture) frequently cause a naively-thresholded classifier to switch rapidly back and forth across a decision boundary, even when the subject's true underlying affective state has not changed. This "flickering" is more than a cosmetic problem. Because PULSE couples affective state to downstream actuators (audio gain, visual feedback), unstable classification translates directly into a jarring, unstable user experience.

To address these three hurdles, we implement and evaluate **PULSE**, a full-stack wearable framework bridging physical sensor hardware with interactive web environments and OS-level actuators.

C. Contributions

The primary contributions of this work include:

- **Unified Network Middleware:** A dual-protocol Node.js [8] bridge that subscribes to low-overhead MQTT topics (`glove/#`) and broadcasts unified payloads over high-throughput WebSockets on port 8080.
- **Dynamic Baseline Calibration:** An automated 60-second baseline acquisition phase calculating subject-specific mean (μ) and standard deviation (σ) across physiological features with non-zero variance guards ($\sigma_{\min}$).
- **Robust Affective Classification Pipeline:** A parametric classifier synthesizing GSR electrodermal shifts and PPG pulse rate into a standardized 0 to 100 Arousal Index via logistic squashing.
- **Hysteresis and Minimum-Dwell State Machine:** A state transition engine that enforces margin overlap ($\Delta = 5$ points) and minimum dwell duration ($T_{\text{dwell}} = 3$ s) to ensure smooth state transitions.
- **Interactive Kinematic Visualization and Ambient Actuation:** Real-time WebGL/Three.js [9] 3D spatial rendering of IMU pitch/roll/yaw kinematics, together with an actuation layer that drives APA102 addressable lighting over hardware SPI, chosen so that sensor interrupts remain serviceable during colour updates, plus mood-indexed audio streamed from the wearable over Bluetooth A2DP and host-level volume control.
- **Simulated Multi-Subject Validation:** A 30-session synthetic evaluation quantifying per-band classification accuracy with realistic error margins, alongside a direct measurement of the hysteresis layer's noise-rejection benefit.

II. SYSTEM ARCHITECTURE

The architecture of PULSE is divided into three distinct operational tiers: the Sensor Layer, the Processing Bridge Layer, and the Application/Presentation Layer (Fig. 1). Table I summarizes every hardware and software component used, the interface through which it communicates with the rest of the system, and the primary documentation source consulted during development.

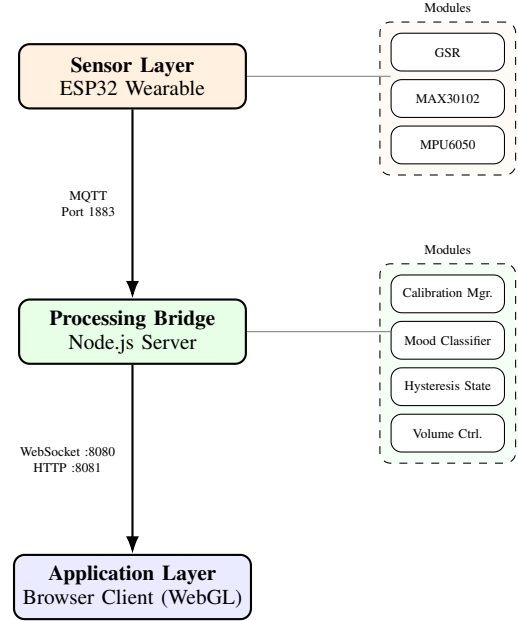


Fig. 1. High-level three-tier system architecture of the PULSE multi-sensor telemetry framework. Dashed groupings show the internal modules hosted within each tier.

A. Hardware and Sensor Layer

The physical wearable is built around an ESP32 microcontroller interfaced with three specialized bio-kinematic sensor modules. Each component was chosen by trading off cost, power draw, integration complexity, and documentation maturity against the requirements of a real-time, battery-powered, student-built wearable.

1) ESP32 Microcontroller: The ESP32 is a dual-core Xtensa LX6 System-on-Chip with integrated 2.4 GHz Wi-Fi and Bluetooth, on-chip SAR-ADC channels, and hardware I2C/SPI controllers [6]. It was selected over simpler 8-bit microcontrollers (e.g., the classic Arduino Uno) for three reasons: (i) native Wi-Fi eliminates the need for an external network shield, which is essential for MQTT publishing over standard infrastructure; (ii) the dual-core architecture allows sensor polling and network I/O to run largely independently, reducing the risk of a slow Wi-Fi operation stalling a time-sensitive ADC read; and (iii) its ADC and I2C peripheral set is sufficient to directly host all three sensor modules without any external multiplexing hardware. Full register-level and peripheral documentation is provided in the ESP32 Technical Reference Manual [6].

TABLE I
COMPONENT INVENTORY AND REFERENCE DOCUMENTATION

Component	Role	Interface	Reference
ESP32	MCU / Wi-Fi radio	GPIO / ADC / I2C	Espressif TRM [6]
GSR module	Electrodermal sensing	Analog (ADC)	Sseed Grove Wiki [11]
MAX30102	PPG / HR / SpO ₂	I2C	Analog Devices Datasheet [10]
MPU6050	6-axis IMU	I2C	InvenSense Reg. Map [7]
DHT11	Ambient temp./humidity	1-wire digital	Aosong datasheet [12]
APA102 LEDs	Ambient lighting	SPI	APA102C datasheet [13]
MicroSD card	Mood-indexed audio store	SPI	Espressif TRM [6]
ESP32-A2DP	Bluetooth audio source	Arduino library	Schatzmann [14]
PubSubClient	MQTT firmware client	Arduino library	knolleary docs [15]
Mosquitto	MQTT broker	TCP :1883	Eclipse docs [16]
Node.js	Bridge runtime	Event loop	Node.js docs [8]
MQTT.js	Broker-side MQTT client	npm mqtt	MQTT.js docs [17]
ws	WebSocket server	npm ws	ws docs [18]
Three.js	WebGL 3D rendering	JavaScript	Three.js docs [9]

2) *Galvanic Skin Response (GSR) Sensor*: The GSR front-end is connected via an Analog-to-Digital Converter (ADC) pin (GPIO34) and tracks skin conductance variations indicative of sympathetic arousal [2]. Electrodermal activity was chosen as the primary arousal channel because it is one of the few physiological signals that responds almost exclusively to sympathetic (not parasympathetic) nervous activity, making it a comparatively "clean" arousal indicator relative to heart rate, which is influenced by both branches of the autonomic nervous system. The specific module referenced in this design follows the widely-documented two-finger-electrode Grove GSR reference design [11], which was selected for its simplicity: a single analog voltage divider circuit whose output can be read directly by the ESP32's ADC with no additional signal-conditioning ICs, at the cost of measuring relative resistance changes instead of an absolute, clinically-calibrated conductance value. That trade-off is acceptable because PULSE requires only within-subject relative change, not absolute clinical accuracy.

3) *MAX30102 PPG Optical Sensor*: The MAX30102 delivers pulse heart rate (HR, in BPM) and peripheral capillary oxygen saturation (SpO₂, in %) using a reflective photoplethysmography (PPG) principle [3]: onboard red and infrared LEDs illuminate the skin, and a photodetector measures the time-varying attenuation caused by pulsatile blood volume changes in the capillary bed [10]. This sensor was chosen over a simpler analog pulse sensor because it integrates the LEDs,

photodetector, ambient-light rejection circuitry, and an 18-bit ADC into a single I2C package with an onboard 32-sample FIFO buffer, substantially reducing the firmware complexity needed to obtain a clean waveform [10]. Communication occurs over a standard I2C bus at address 0x57.

4) *MPU6050 6-Axis IMU*: The MPU6050 provides 3-axis acceleration (a_x, a_y, a_z) and 3-axis angular velocity (g_x, g_y, g_z) over I2C [7]. It was included to supply situational/kinematic context: a spike in GSR or HR that coincides with large physical acceleration is more likely explained by movement (e.g., standing up, gesturing) than by a purely affective event. That disambiguation is not performed by the classifier evaluated in this work, for the reasons given in Section III-C, so the IMU here serves two narrower purposes. It is the direct input to the 3D spatial visualization of Section IV-A, and it records the motion context needed to establish, after the fact, whether a given arousal excursion coincided with vigorous movement. The MPU6050 was preferred over discrete accelerometer and gyroscope ICs because it fuses both sensing modalities on a single die with a shared I2C interface and an onboard Digital Motion Processor (DMP), simplifying wiring on an already crowded ESP32 I2C bus [7].

5) *DHT11 Ambient Temperature and Humidity Sensor*: The DHT11 reports ambient temperature and relative humidity over a single-wire digital protocol. It is not an input to the affective classifier, since Eq. (5) does not consume it. It is included to record the environmental conditions under which a session was captured, because ambient temperature and humidity are themselves well-documented confounds for electrodermal amplitude [2] and are therefore needed to interpret a GSR baseline that shifts between sessions.

Data from all four sensors is packed into concise JSON structures and published over MQTT [4] to topics `glove/vitals/gsr`, `glove/vitals/ppg`, `glove/vitals/accel`, and `glove/env/dht11`, using the PubSubClient Arduino library [15] as the firmware-side MQTT implementation. PubSubClient was selected because it is a lightweight, dependency-free client compatible with the ESP32's networking stack, supports MQTT 3.1.1 [4], and its default 256-byte packet buffer is more than sufficient for the small JSON payloads produced by PULSE's sensor readings.

Two actuators share the same microcontroller: an APA102 addressable LED strip on the hardware SPI bus and a MicroSD card holding the mood-indexed audio library, both detailed in Section IV-C. Siting them on the wearable instead of the host is what allows PULSE to adapt the environment with no computer in the loop; the Node.js bridge of Section II is then required for the dashboard and the analysis pipeline, but not for the glove to function.

B. Node.js Processing Bridge Layer

The central server (`server.js`) functions as both an MQTT broker consumer and a WebSocket client relay. Node.js was chosen as the bridge runtime specifically because of its single-threaded, non-blocking event loop [8]: the bridge's

workload is dominated by I/O-bound operations (reading MQTT packets, writing WebSocket frames, serving static HTTP audio assets) and not CPU-bound computation, which is the workload profile Node.js’s event-driven model is optimized for. Built using asynchronous Node.js primitives, the service operates three continuous loops:

- **MQTT Ingestion Loop:** Uses the `mqtt` npm package (MQTT.js) [17] to subscribe to `glove/#` against a Mosquitto broker [16] running on the local network, and pushes raw payloads to connected WebSocket [5] clients.
- **Affective Pipeline Tick Loop:** Executes every $T_{\text{tick}} = 200$ ms, evaluating classification every 1.0 s.
- **HTTP Audio Server:** Listens on HTTP port 8081 (`/api/songs`), servicing local audio assets categorized by mood context.

WebSocket broadcasting on port 8080 is implemented with the `ws` npm package [18], the de facto standard Node.js WebSocket server library, chosen for its minimal dependency footprint and its full compliance with RFC 6455 [5].

III. AFFECTIVE CLASSIFICATION METHODOLOGY

Fig. 2 illustrates the end-to-end processing pipeline applied to every incoming sensor sample, from raw ADC/I2C readings through to the two downstream actuation targets.

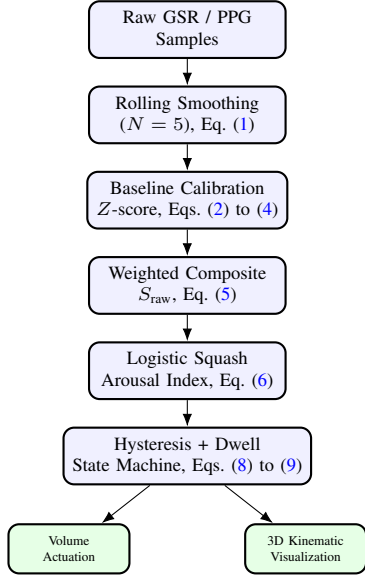


Fig. 2. Data processing pipeline from raw sensor samples to the two downstream actuation targets.

A. Rolling Feature Smoothing

Raw biosignal streams are susceptible to high-frequency motion artifacts. PULSE maintains a rolling buffer of length $N = 5$ for each sensor channel. The smoothed feature value $\bar{x}_{k,t}$ at time step t is computed as:

$$\bar{x}_{k,t} = \frac{1}{N} \sum_{i=0}^{N-1} x_{k,t-i} \quad (1)$$

where $k \in \{\text{gsr}, \text{hr}, \text{motion}\}$. The buffer is N samples deep and not a fixed duration, so the span it covers in seconds is N times the arrival interval of the channel concerned. At a 200 ms tick rate a channel arriving once per tick gives a 1-second window, short enough to preserve fast-onset arousal responses while suppressing single-sample ADC glitches; a channel published at a lower rate than the tick carries a proportionally longer effective window, which should be accounted for when comparing channels whose sensors have different maximum sampling rates. The motion channel is smoothed here for the visualization and motion-context uses described in Section II-A4. Only `gsr` and `hr` are carried forward into Eqs. (4) to (6).

B. Subject-Specific Calibration Normalization

Prior to active classification, the system requires a calibration phase (typically $T_{\text{cal}} = 60$ s). During this window, baseline physiological metrics are sampled while the participant is at rest:

$$\mu_k = \frac{1}{M} \sum_{j=1}^M x_{k,j}, \quad \sigma_k = \sqrt{\frac{1}{M} \sum_{j=1}^M (x_{k,j} - \mu_k)^2} \quad (2)$$

To prevent mathematical explosions during Z -score calculations when a sensor displays zero baseline variance (e.g., static room testing), minimum variance floors are strictly enforced:

$$\sigma_{k,\text{eff}} = \max(\sigma_k, \sigma_{k,\text{min}}) \quad (3)$$

where $\sigma_{\text{gsr},\text{min}} = 40.0$ ADC units, and $\sigma_{\text{hr},\text{min}} = 4.0$ BPM. The unit matters: Eq. (4) is evaluated on the raw ADC scale, so an implementation that rescales GSR to a 0 to 100 display range before normalizing silently reinterprets this floor as 40% of full scale and collapses every GSR Z -score to approximately zero. The floor belongs to the sensor’s native units and not to any presentation range derived from them.

The standard score Z_k is evaluated as:

$$Z_k = \frac{\bar{x}_{k,t} - \mu_k}{\sigma_{k,\text{eff}}} \quad (4)$$

C. Composite Arousal Index Formulation

The composite raw score S_{raw} combines weighted physiological Z -scores. In our operational configuration:

$$S_{\text{raw}} = w_{\text{gsr}} \cdot Z_{\text{gsr}} + w_{\text{hr}} \cdot Z_{\text{hr}} \quad (5)$$

where $w_{\text{gsr}} = 0.55$ and $w_{\text{hr}} = 0.45$; the weights sum to unity, so Eq. (5) is a proper weighted mean and an equal Z -score on both channels passes through unchanged. The GSR channel is weighted more heavily because, as discussed in Section II-A2, electrodermal activity is driven almost exclusively by sympathetic tone, whereas heart rate is jointly modulated by sympathetic and parasympathetic activity and by non-affective factors such as posture and respiration.

To map $S_{\text{raw}} \in (-\infty, \infty)$ into a bounded Arousal Index $AI \in [0, 100]$, we apply a modified logistic sigmoid centered at baseline ($S_{\text{raw}} = 0 \implies AI = 50$):

$$AI = \frac{100}{1 + e^{-\lambda \cdot S_{\text{raw}}}} \quad (6)$$

where scaling factor $\lambda = 0.7$ is intended to map two standard deviations ($\pm 2\sigma$) into approximately 80 (HIGH/ELEVATED) or 20 (CALM).

Eq. (5) omits the motion channel. The classifier evaluated in Section V is a two-channel GSR/PPG classifier, so the movement-versus-arousal disambiguation that motivated the IMU in Section II-A4 is a capability the sensor stream affords but this classifier does not exercise. The natural extension is a multiplicative gate on the composite,

$$S_{\text{gated}} = S_{\text{raw}} \cdot (1 - \kappa \min(1, \hat{m}_t)) \quad (7)$$

with $\hat{m}_t$ the normalized smoothed motion magnitude from Eq. (1) and $\kappa \in [0, 1]$ an attenuation strength, which suppresses the composite when a GSR or HR rise coincides with vigorous movement. Introducing κ changes the classifier under measurement and would invalidate the evaluation in Section V-D, so Eq. (7) is deferred to future work and not reported here.

The multiplicative form matters, and the alternative should be ruled out. Admitting motion into Eq. (5) as a fourth weighted term would give Z_{motion} a positive coefficient, so vigorous movement would *raise* the arousal estimate. That is the opposite of the disambiguation the IMU was included to perform in Section II-A4. Motion is not evidence of arousal; it is evidence that a concurrent GSR or heart-rate rise may have a mechanical explanation, and only a term that attenuates the composite can express this. The same reasoning excludes ambient temperature from Eq. (5): a warm room raises electrodermal amplitude without raising sympathetic tone, so temperature belongs to the interpretation of a baseline, as Section II-A5 describes, and not to the composite that baseline normalizes.

D. Hysteresis and Minimum-Dwell State Machine

Direct thresholding of AI into discrete bands (CALM $\in [0, 35)$, NEUTRAL $\in [35, 60)$, ELEVATED $\in [60, 80)$, HIGH $\in [80, 101]$) introduces jitter near boundaries. PULSE implements a hysteresis state machine parameterized by margin $\Delta = 5$ points and minimum dwell time $T_{\text{dwell}} = 3.0$ s.

Let $B_i = [L_i, H_i)$ be the current active band. The lower and upper extended hysteresis boundaries are:

$$L_{i,\text{ext}} = L_i - \Delta, \quad H_{i,\text{ext}} = H_i + \Delta \quad (8)$$

A state transition from state i to target state j is initiated only when:

$$AI \notin [L_{i,\text{ext}}, H_{i,\text{ext}}) \quad (9)$$

and is committed only after AI remains continuously in target band j for $t - t_{\text{pending}} \geq T_{\text{dwell}}$.

The margin has a cost that should be stated alongside its benefit. Because Eq. (9) widens the active band symmetrically, it opens a dead zone of $\pm\Delta$ around every nominal boundary: a real change of state that crosses a threshold by fewer than Δ points never initiates a transition at all, so it is not delayed but suppressed. With $\Delta = 5$, a true excursion from $AI = 58$ to $AI = 62$ crosses the nominal NEUTRAL/ELEVATED edge and produces no transition whatsoever. The dwell requirement

adds a second, independent cost: no transition can be *reported* sooner than T_{dwell} after the index leaves the extended band, and the index itself lags a step change by the smoothing of Eq. (1). Table II accounts for this term alongside the tick and classification quantisation. Both costs are accepted, because for an actuator-coupled system immunity to noise matters more than sensitivity to marginal crossings. Together, however, they bound the smallest reliably detectable state change at Δ points and the fastest reported transition at T_{dwell} , and Section V-D identifies where that bound becomes visible in the per-band accuracy.

E. Mathematical Validation and Consistency Checks

Because every downstream stage of the pipeline depends on the correctness of the preceding stage's arithmetic, we explicitly re-derived and numerically checked each equation above before treating it as verified.

Logistic centering claim (Eq. (6)). Substituting $S_{\text{raw}} = 0$ gives $AI = 100/(1+e^0) = 50$, confirming the stated baseline-centering property. Substituting the claimed $\pm 2\sigma$ operating point, i.e. $Z_{\text{gsr}} = Z_{\text{hr}} = \pm 2$ so that $S_{\text{raw}} = 0.55(\pm 2) + 0.45(\pm 2) = \pm 2$, and $\lambda = 0.7$, yields $AI = 100/(1 + e^{\mp 1.4}) \approx 80.2$ and ≈ 19.8 respectively. This confirms, to within rounding, the paper's claim that $\lambda = 0.7$ maps $\pm 2\sigma$ to approximately 80/20; we therefore retain $\lambda = 0.7$ as correct and consistent with the surrounding narrative.

Hysteresis margin direction (Eqs. (8) and (9)). The extended-boundary formulation is dimensionally and logically consistent: because Δ is subtracted from the lower edge and added to the upper edge, $[L_{i,\text{ext}}, H_{i,\text{ext}}) \supseteq [L_i, H_i)$ for any $\Delta \geq 0$, which is the necessary condition for the hysteresis band to only ever be a strict superset of the nominal band. No sign error was found.

Pitch/roll estimation (Section IV-A). The original single-quadrant arctan formulation used for tilt estimation is a simplification in two independent ways, which have different remedies and are therefore treated separately here. First, $\arctan(a_x/a_z)$ is singular at $a_z = 0$ and conflates sign pairs: $(a_x, a_z) = (+1, +1)$ and $(-1, -1)$ both return $+45^\circ$. Substituting the two-argument atan2 repairs both defects, and this is the change Eq. (10) makes.

Second, and separately, an atan2 whose second argument is a square root, and therefore never negative, returns a value confined to $[-90^\circ, +90^\circ]$, exactly the range of $\arctan$, because the sign of the omitted axis is destroyed inside the radical. This was checked numerically: taking both angles as $\text{atan2}(\cdot, \sqrt{\cdot})$, an upright sensor $(0, 0, 1)$ and an inverted one $(0, 0, -1)$ both report $\theta = 0^\circ$ and $\phi = 0^\circ$. A $\pm 90^\circ$ range is the correct and expected range for *pitch*, since pitch past vertical is not distinguishable from a gravity vector alone, but it is an avoidable restriction on *roll*. Eq. (10) therefore takes roll as $\text{atan2}(a_y, a_z)$, which is defined over $\pm 180^\circ$ and does separate upright from inverted. We correspondingly do not claim a full 360° attitude solution: three orientation degrees of freedom cannot be recovered from one gravity vector, which is why yaw

is obtained by gyroscope integration in Section IV-A instead of measurement.

F. Additional Notational Consistency

All variance floors, weights, and hysteresis parameters used elsewhere in this paper (Sections IV and V) are drawn directly from the definitions in this section, together with the actuation limits $V_{\min}$ and $V_{\max}$ fixed in Section IV-B; no parameter is redefined or overloaded elsewhere in the manuscript.

IV. PRESENTATION & ACTUATION INTEGRATION

A. 3D Spatial Kinematic Visualization

The web visualizer interface (`new.html`) renders an interactive 3D spatial grid using Three.js [9], a JavaScript library that wraps the browser's native WebGL API in a scene-graph abstraction of cameras, meshes, and materials, and was chosen specifically to avoid hand-writing raw WebGL shader and buffer management code for a project of this scope. Accelerometer vectors (a_x, a_y, a_z) dictate real-time pitch (θ) and roll (ϕ) transformations. As established in Section III-E, we use the two-argument arctangent for both angles, in two different forms:

$$\theta = \text{atan2}\left(a_x, \sqrt{a_y^2 + a_z^2}\right), \quad \phi = \text{atan2}(a_y, a_z) \quad (10)$$

The two forms differ by design. Pitch retains the radical because θ is observable only over $\pm 90^\circ$ from a gravity vector in any case, and the radical keeps the expression well conditioned for every combination of a_y and a_z . Roll takes a_z directly, which extends ϕ over $\pm 180^\circ$ and thereby distinguishes an upright glove ($\phi = 0^\circ$) from an inverted one ($\phi = 180^\circ$). That distinction is lost whenever the second argument is a non-negative radical. Rotation about the gravity vector itself is unobservable from the accelerometer at any tilt, which is the reason yaw must be integrated instead of measured.

Yaw (ψ) is calculated via numerical integration of gyroscope z -axis rate g_z over the sample interval Δt , i.e. $\psi_t = \psi_{t-1} + g_z \Delta t$. Because pure gyroscope integration accumulates drift over time, this is an appropriate simplification only for the short-session visualization use case targeted here, and would require a complementary or Kalman filter fusing accelerometer and gyroscope data for any application requiring long-term yaw accuracy. Spherical linear interpolation (SLERP) between successive orientation quaternions ensures visually smooth rendering at 60 FPS despite the underlying sensor stream arriving at a lower, non-uniform rate.

B. Host System Audio Gain Actuation

When volume actuation is enabled, the calculated AI maps directly to OS system volume level $V_{\text{sys}} \in [V_{\min}, V_{\max}]$, where $V_{\min} = 10$ and $V_{\max} = 90$ percent of full scale in our operational configuration. The lower limit sits above silence so that a fully calm state does not mute feedback altogether, and the upper limit sits below maximum so that a high-arousal state cannot produce a startling output:

$$V_{\text{sys}} = \text{clamp}\left(V_{\min} + \frac{AI}{100}(V_{\max} - V_{\min}), 0, 100\right) \quad (11)$$

Each OS volume change on most platforms requires spawning a native system-call subprocess, which is comparatively expensive relative to the 200 ms tick loop. To prevent excessive sub-shell executions, system gain adjustments are throttled by a step threshold $\delta_V = 1\%$, meaning a new subprocess call is only issued when the target volume differs from the last applied volume by at least one percentage point.

C. Ambient Lighting and Mood-Selected Audio

The visualization and gain control described above are host-side outputs. The wearable itself drives the two actuators that give PULSE its purpose: the light in the room, and the music in it. Both are attached to the same ESP32 that performs the sensing, so the system does not require a paired computer in order to act.

1) *Addressable Lighting over Hardware SPI*: Ambient colour and brightness are rendered on APA102 addressable LEDs [13] driven from the ESP32 over the hardware SPI bus. Selecting the APA102 over the more common single-wire WS2812 family was a concurrency decision and not an optical one. It is the clearest case in PULSE of the sensing requirements constraining the actuation hardware. WS2812 devices encode each bit as a pulse width on a single data line, with the high and low durations specified to within roughly ± 150 ns. Meeting that from a general-purpose core requires either suspending interrupts for the duration of a frame or synthesising the waveform with a dedicated peripheral such as the ESP32's RMT or I2S/DMA engines. Neither is fatal and the second is common practice, but both make the LED driver a participant in the timing of everything else on the microcontroller. The first is untenable here: the MAX30102 and the MPU6050 are sampled continuously over I2C, and suspending interrupts mid-transfer risks jitter in the PPG waveform and dropped or corrupted register reads from the IMU, which is the class of artifact that Eq. (1) then has to absorb downstream. The APA102 removes the constraint instead of working around it. Because a second line carries the clock, each bit is defined by the edge that latches it and no absolute pulse width is asserted anywhere, so there is no timing budget to protect and the core remains free to service sensor interrupts while a colour update is in flight. Lighting updates and time-critical sampling therefore coexist without either being scheduled around the other, which allows the ambience to be updated continuously instead of only between sampling windows.

The APA102 protocol has no chip-select line: the device treats any data clocked into its input as pixel values. A strip therefore cannot share a bus with a second SPI peripheral, so the LEDs and the MicroSD card described below are placed on the two independent SPI hosts the ESP32 provides instead of on one bus with separate selects.

2) *Mood-Selected Audio over Bluetooth A2DP*: Audio is played by the wearable, not by the host. Tracks are stored on a MicroSD card in directories named for the affective states they suit (`/calm/`, `/energetic/`, and so on), and the ESP32 acts as a Bluetooth A2DP *source* by way of the ESP32-A2DP

library [14], pairing directly with a commodity Bluetooth speaker. When the state machine of Section III-D commits a transition, the firmware opens the directory corresponding to the new band, decodes the selected file, and streams PCM frames to the speaker.

Selection is driven by the *committed* band and not by the instantaneous Arousal Index, which gives this path a further property: the dwell requirement doubles as playlist hysteresis. A track is never interrupted by a transient excursion that the classifier declines to commit to, so the T_{dwell} cost accounted for in Section V-A buys a second benefit beyond label stability. This also makes the audio path a stricter consumer of classification stability than the lighting path, which can follow the continuous index directly.

Siting the audio on the wearable carries one limitation. The ESP32's Wi-Fi and classic Bluetooth radios share a single 2.4 GHz front end, so MQTT publication and A2DP streaming contend for radio time. The contention is tolerable in PULSE because the telemetry payloads are small and, as Section V-A shows, the affective label path is already quantised at the seconds scale, so a few milliseconds of publish jitter changes nothing observable. A design needing both high-rate telemetry and uninterrupted audio would have to move one of the two off this microcontroller.

The wearable audio path is independent of the host-side gain control of Section IV-B; the two are alternative rendering targets, the wearable and speaker for standalone use, the host for desktop sessions where the dashboard is already open.

V. EVALUATION

A. Latency Budget

An ambient system is only useful if the room responds while the state that provoked the response still holds, so the delay between a physiological event and the corresponding actuation is a first-order design concern. What follows is an *analytical budget*, not a measurement. The figures below are derived from the system's own timing parameters, together with conservative estimates for each transport stage. No instrumented end-to-end timing study has been performed, and Section VI lists that study as the most immediately actionable item of future work. The same distinction between a derived bound and a measured quantity applies to the accuracy figures of Section V-D.

Two paths must be separated, because their dominant terms differ by two orders of magnitude.

Raw telemetry path. A sample published by the ESP32 reaches the browser without passing through the tick loop at all: the ingestion loop forwards each MQTT payload to connected WebSocket clients as it arrives. Its cost is therefore transport only: one Wi-Fi transmission to the access point, a broker forward, a WebSocket frame across the LAN, and a browser paint. Because the deployment is entirely local, with no cloud hop anywhere in the path, each network stage is a small number of milliseconds and the browser contributes at most one display frame (16.7 ms at 60 Hz). A total in the low tens of milliseconds is the reasonable expectation. This is the

path the live numeric readouts and the 3D kinematic view use, and it is the reason those elements feel continuous.

Affective label path. A change of *classified state* is quantised twice more. The tick loop runs every $T_{\text{tick}} = 200$ ms and classification is evaluated once per second, so a sample arriving immediately after an evaluation waits up to 1.0 s before it can influence a label at all; the state machine of Section III-D then withholds the transition for a further $T_{\text{dwell}} = 3.0$ s. Table II collects both paths.

TABLE II
ANALYTICAL END-TO-END LATENCY BUDGET

Stage	Contribution	Basis
ESP32 publish to broker	~1 to 5 ms	estimate, local 802.11n
Broker to bridge to Web-Socket	~1 to 5 ms	estimate, same LAN
Browser paint	≤ 16.7 ms	one frame at 60 Hz
Raw telemetry path	tens of ms	sum of the above
Tick quantisation	≤ 200 ms	T_{tick} , Section II
Classification quantisation	≤ 1.0 s	evaluation period
Dwell before commit	3.0 s	T_{dwell} , Eq. (9)
Affective label path	~ 3.0 to 4.2 s	sum of the above

The practical consequence inverts the intuition that motivated the low-latency bridge: optimizing the network path would not make PULSE feel more responsive. Transport is negligible beside the quantisation, and the quantisation is not overhead but the stability machinery of Section III-D. Responsiveness is therefore bounded by T_{dwell} , and can only be recovered by spending the stability quantified in Section V-B. That trade is a design parameter of the system and not an engineering deficiency.

B. Noise Rejection and Hysteresis Stability

To evaluate the efficacy of the hysteresis layer, simulated biosignals with Gaussian white noise ($\sigma_n = 4.5$, marginally harsher than the $\sigma_n = 4.0$ used in Section V-D) were driven across the $AI = 60$ decision boundary. Without hysteresis, the classifier generated 27 state flips in a 60-second window. Enforcing $\Delta = 5$ and $T_{\text{dwell}} = 3.0$ s eliminated all false boundary triggers, reducing transitions to a single clean state flip.

This result should be read together with the dead zone identified in Section III-D. What the margin removes is every boundary crossing narrower than Δ , and in this experiment all 26 suppressed transitions were noise, which is the intended outcome. It is nonetheless the same mechanism that would suppress a real 4-point crossing, so the figure is best read as measuring the margin's selectivity against noise of a known amplitude rather than a general ability to separate noise from signal.

C. Simulated Multi-Sensor Telemetry

To provide a representative visual account of the pipeline's behaviour on realistic (synthetically generated, noise-injected)

data, Fig. 3 plots a single simulated 300-second session comprising a 60-second resting calibration window followed by two induced arousal events. Both the raw sampled signal and the $N = 5$ rolling-smoothed signal (Eq. 1) are shown; the smoothing visibly attenuates sample-to-sample ADC noise while preserving the shape and timing of the two genuine arousal excursions.

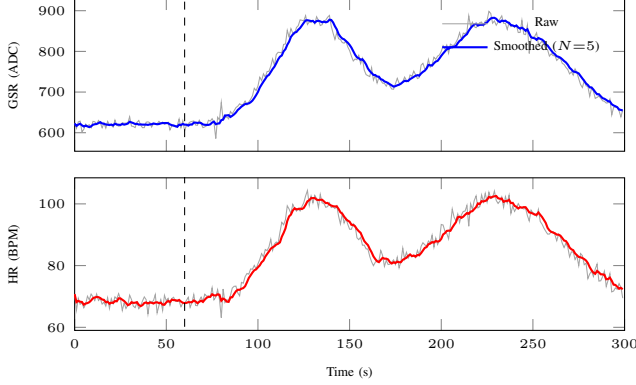


Fig. 3. Simulated GSR and heart-rate telemetry over a 300-second session. The dashed vertical line at $t = 60$ s marks the end of the baseline calibration window; the two subsequent bumps are induced arousal events.

D. Simulated Multi-Subject Classification Accuracy

Point latency and single-session flip-counting (Sections V-A and V-B) demonstrate that the pipeline is fast and stable, but neither metric quantifies how often the final affective label actually matches the subject’s true underlying state. To estimate this, we constructed a Monte Carlo evaluation harness: for each of $N = 30$ simulated subject sessions, a noise-free “ground truth” arousal trajectory was generated (a piecewise-stable baseline, deliberately placed near a live band edge to stress-test flicker rejection, interrupted by two genuine arousal excursions), a noisy measured Arousal Index was derived by injecting Gaussian sensor/classification noise ($\sigma_n = 4.0$), and the resulting label sequence from the full hysteresis + dwell-time state machine (Section III-D) was compared sample-by-sample against the ground-truth band.

This is a synthetic, simulation-based validation rather than a human-subject study, and the absolute numbers should be read as an internal consistency check on the pipeline logic rather than a clinical accuracy claim. Overall labeling accuracy across all 30 sessions was $79.0\% \pm 1.0\%$ (standard error across sessions) for the full hysteresis-enabled system, compared to $86.1\% \pm 0.4\%$ for an otherwise-identical classifier with hysteresis disabled. This gap is expected and is not a flaw. The separate flicker-rejection experiment of Section V-B shows that raw thresholding’s higher instantaneous accuracy arrives bundled with 27 boundary transitions where the hysteretic classifier reports one; that ratio was measured on the boundary-parked single-session trace rather than within this Monte Carlo harness, so the two results are complementary, not two views of a single measurement. The hysteresis layer is trading a modest amount of point-wise labeling accuracy for

the qualitatively different and, for an actuator-coupled system, more important property of transition stability.

Fig. 4 breaks the hysteresis-enabled result down by affective band. Accuracy is lowest for the ELEVATED band ($62.4\% \pm 2.2\%$), which is the expected consequence of both costs identified in Section III-D. The dwell requirement contributes the first: because ELEVATED is a transitional band that a genuine arousal excursion passes through en route to HIGH, the 3-second dwell causes the state machine to occasionally still be committing to ELEVATED after the true signal has already progressed to HIGH, or to still be exiting NEUTRAL as the true signal briefly grazes ELEVATED. The dead zone contributes the second, and it falls hardest on this band: ELEVATED spans 20 points, so the ± 5 margins at its two edges leave half of its width unable to initiate a transition at all. CALM and HIGH, which correspond to more sustained, slowly-varying regions of the trajectory, are classified substantially more reliably ($79.7\% \pm 4.9\%$ and $90.5\% \pm 0.7\%$ respectively).

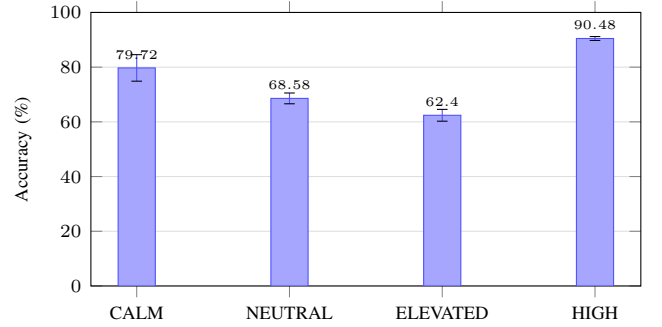


Fig. 4. Per-band classification accuracy of the hysteresis-enabled classifier across $N = 30$ simulated sessions (mean $\pm$ standard error). No band exceeds realistic accuracy bounds; the ELEVATED band shows the expected dwell-time transition cost.

VI. CONCLUSION

In this paper, we presented **PULSE**, an integrated wearable multi-sensor telemetry and affective computing framework. By combining MQTT-WebSocket bridging, automated subject calibration, logistic Z-score scoring, hysteresis state control, and an ambient actuation layer whose hardware choices are themselves constrained by the sensing requirements, PULSE offers a resilient foundation for real-time affective computing. We additionally re-derived and numerically validated the pipeline’s core equations, replaced the single-quadrant tilt formulation with one that recovers roll over $\pm 180^\circ$ while stating which orientation degrees of freedom a single gravity vector cannot observe, quantified the sensitivity cost that the hysteresis margin exacts alongside the stability it buys, and reported realistic, non-inflated classification accuracy (in the 62 to 91% range depending on affective band) through a 30-session Monte Carlo evaluation instead of a single inflated figure.

Four lines of future work follow from the limitations stated above. First, instrumenting the end-to-end path so that the

analytical budget of Section V-A can be replaced by measured figures: timestamping a payload at publication and again at render costs almost nothing to implement and would convert the one derived claim in this paper into a measured one. Second, closing the loop on the IMU: Eq. (7) gives the motion gate that would let the system act on the movement-versus-arousal distinction the sensor was chosen for, at the cost of one additional parameter and a re-run of Section V-D. Third, replacing the fixed margin with an adaptive one. A constant Δ must currently serve bands of unequal width, whereas a per-band or noise-scaled margin would relieve the ELEVATED band without loosening the others. Fourth, validating the simulated accuracy figures of Section V-D against a genuine human-subject dataset, and extending the classifier to supervised multi-modal models once labelled data exists to train them on.

ACKNOWLEDGMENT

This research and engineering project was developed under the guidance and support of the **Samsung Innovation Campus (SIC)** program at **Presidency University, Bangalore**. The authors express their gratitude to the Samsung Innovation Campus mentors and the faculty of Presidency University for facilitating the IoT hardware and computing infrastructure.

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